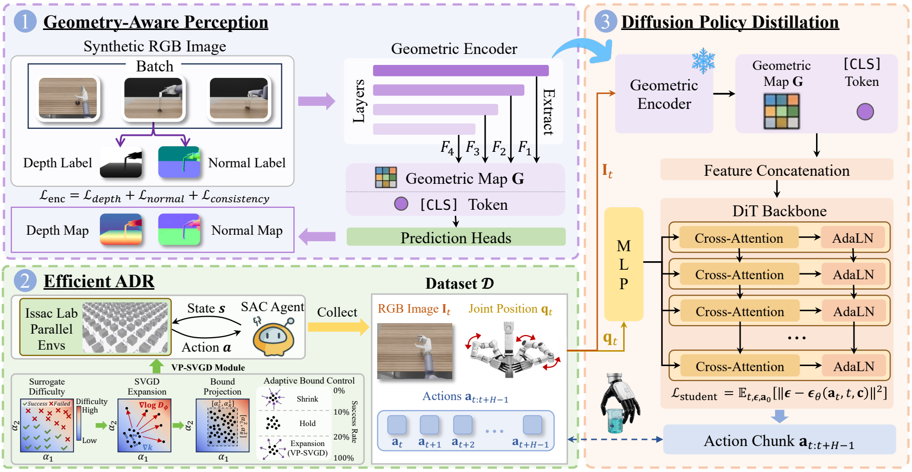

Fig. 1: Overview of GEARS. Stage 1: A geometric encoder is fine-tuned with depth and surface normal supervision on synthetic data, producing domain-invariant geometric features; the prediction heads are discarded after training. **Stage 2**: A privileged RL expert is trained in massively parallel environments with VP-SVGD, which steers domain randomization toward difficult regions for efficient learning. **Stage 3**: The expert is distilled into a diffusion policy that conditions on the frozen geometric encoder, generating temporally coherent action chunks for real-world deployment.

On the control side, we introduce VP-SVGD, a particle-based domain randomization method that steers the training distribution toward difficult regions rather than expanding all parameter ranges uniformly as in standard ADR [1]. This produces a robust RL expert more sample-efficiently. We then distill the expert into a diffusion policy because domain randomization induces multimodal action distributions [7]: the same observation can correspond to different optimal actions under different physical parameters. The diffusion formulation captures this full distribution and produces temporally coherent action sequences, avoiding the averaging artifacts of unimodal alternatives.

3.1 Geometry-Aware Perception

Unlike textures and lighting, geometric properties such as depth and surface orientation transfer naturally between simulation and reality [13, 36]. We exploit this invariance by fine-tuning a pretrained visual foundation model to jointly predict depth and surface normals from monocular RGB, using the privileged annotations that simulation provides for free. The joint prediction, coupled with a cross-task consistency constraint, forces the encoder to internalize the 3D structure of the scene rather than relying on domain-specific appearance cues.

Architecture. We build upon Depth Anything V2 Small [37], which pairs a DINOv2 ViT backbone with a DPT fusion neck [30]. Given an RGB image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, the ViT backbone extracts multi-scale features $\{\mathbf{F}_i\}_{i=1}^4$ from four